

NeurIPS 2026 • Evaluations & Datasets Track

RASS

Risk-Audited Budget Selection for Compact NeRF Benchmark Subsets

48 scenes instead of 3,521, backed by a statistical audit

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Full NeRF benchmarks are too costly to rerun for every idea

3,521

NeRF trainings for every ablation on the full benchmark

512

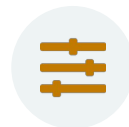
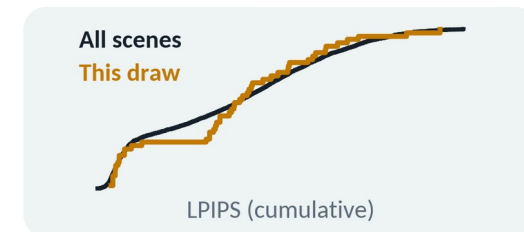
H100 GPUs, at most, to produce the full logs once

Ad hoc subsets can look right for the wrong reasons



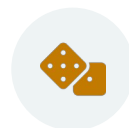
Tail shift

A real 48-scene draw: every mean matches, but its LPIPS distribution (amber) drifts from the full set



Metric leakage

Another real draw: PSNR uses 1% of its tolerance, while SSIM and LPIPS exceed theirs



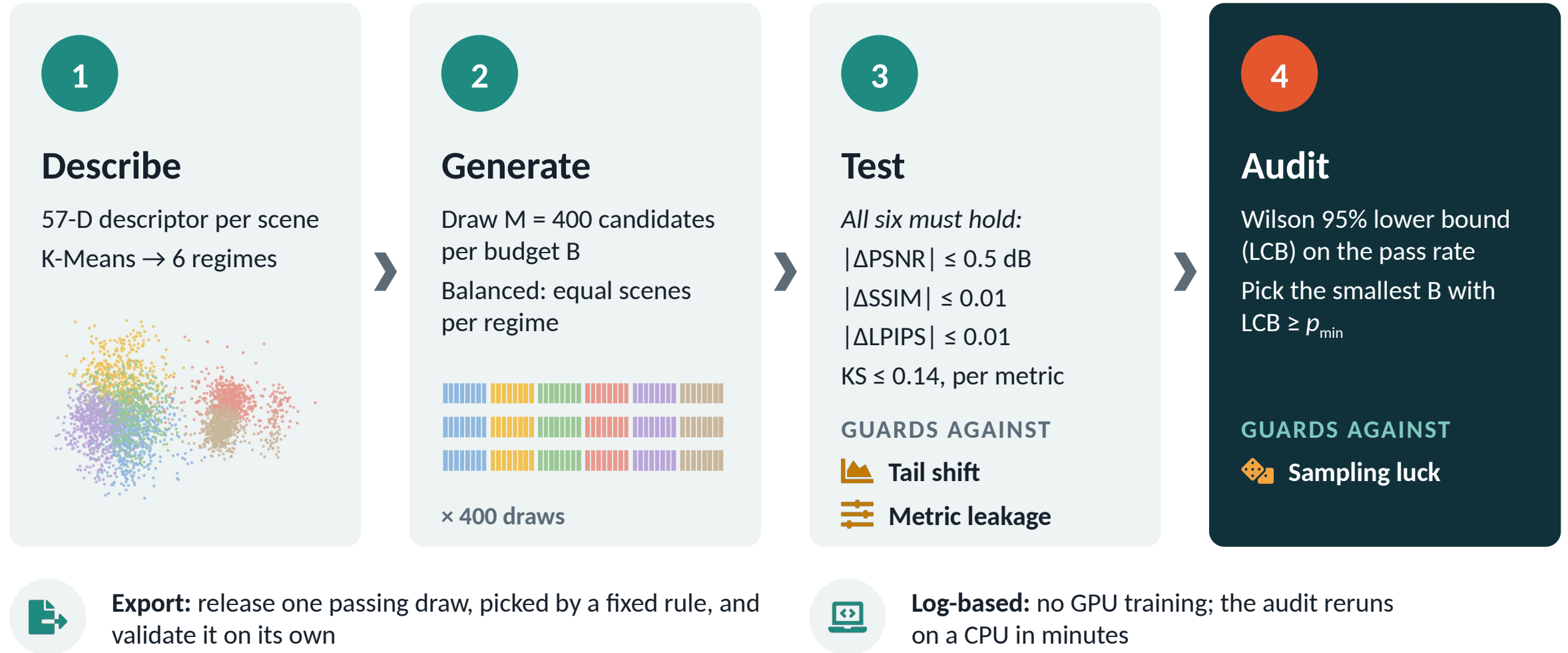
Sampling luck

One subset passes by chance; the next draw does not

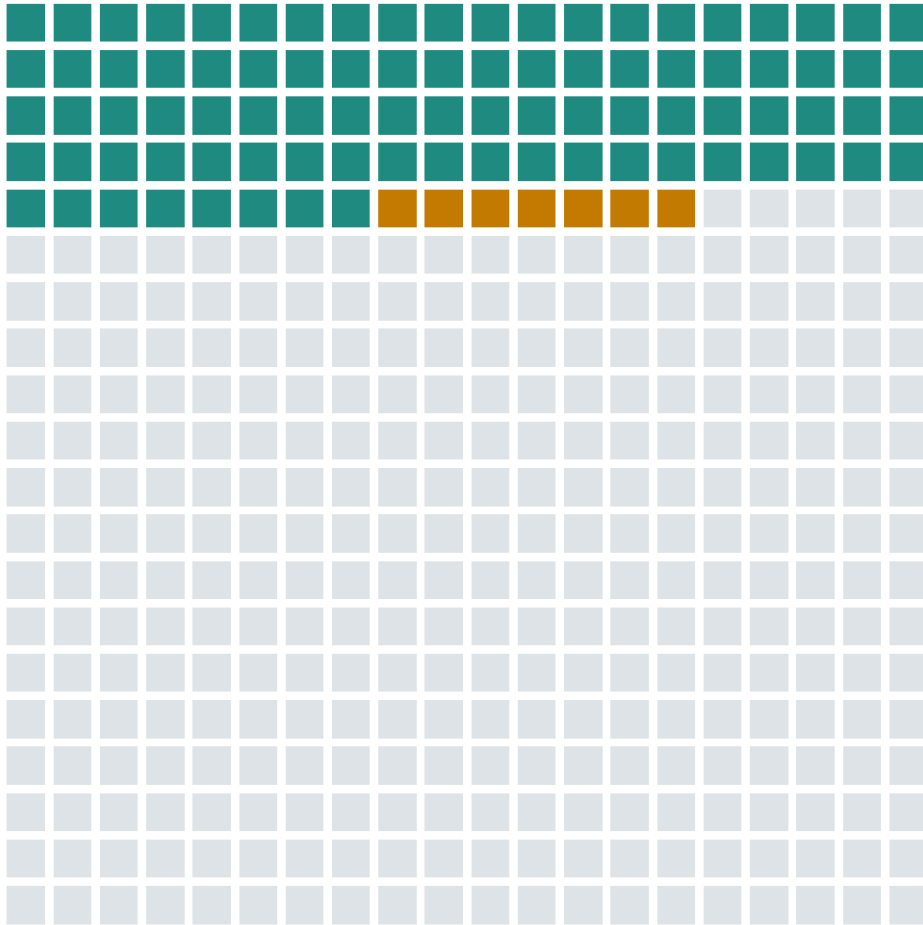


Our question: what is the smallest subset that preserves the full-benchmark summaries, under explicit fidelity constraints?

RASS audits a subset-generation procedure, not one lucky subset



At 48 scenes, even the pessimistic pass rate clears the target



- Pass all six constraints (88)
- Means match, KS fails: tail shift caught (7)
- Fail a mean constraint (305)

Pass rate $88 / 400 = 0.22$

Wilson 95% LCB $0.182 \geq p_{\min} = 0.08$

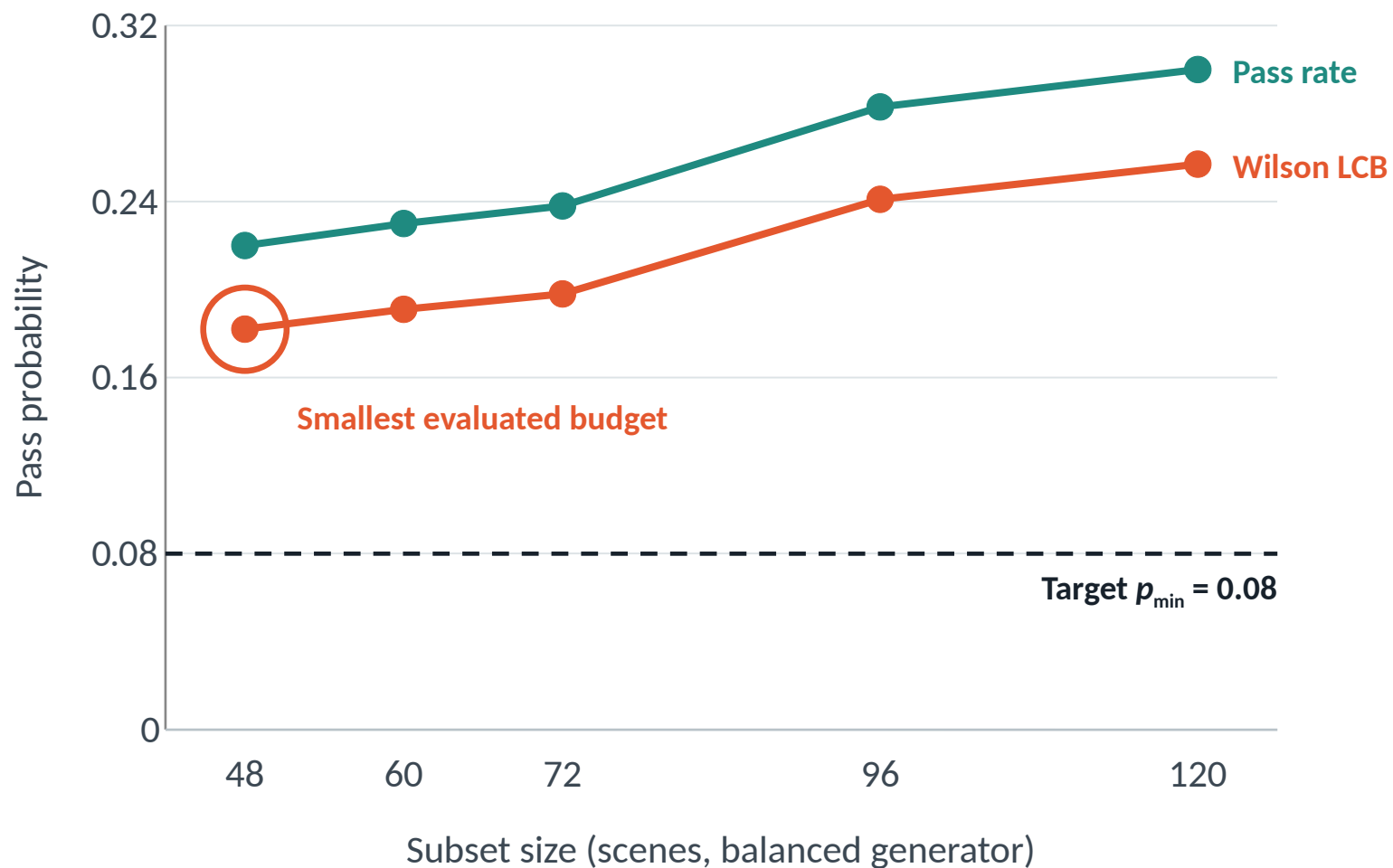


48 scenes certified

$p_{\min}$ is a yield: on average, at most 12.5 cheap log checks find a passing subset.

The 400 real candidate draws at 48 scenes

The frontier trades scenes for reliability



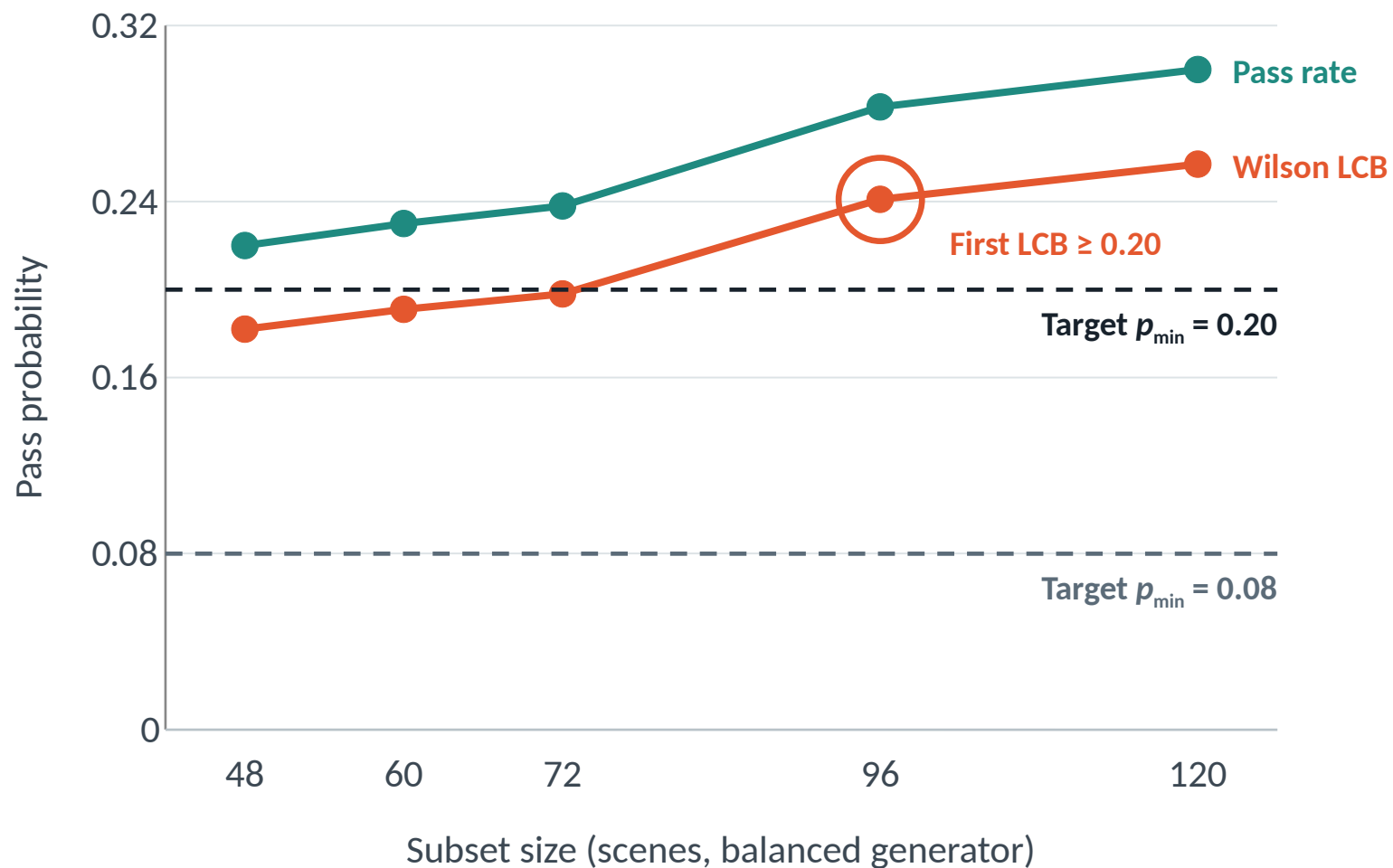
Target $p_{\min} = 0.08$

RASS-48

48 scenes · LCB 0.182

Screening & ablations

The frontier trades scenes for reliability



Target $p_{\min} = 0.08$

RASS-48

48 scenes · LCB 0.182

Screening & ablations

Target $p_{\min} = 0.20$

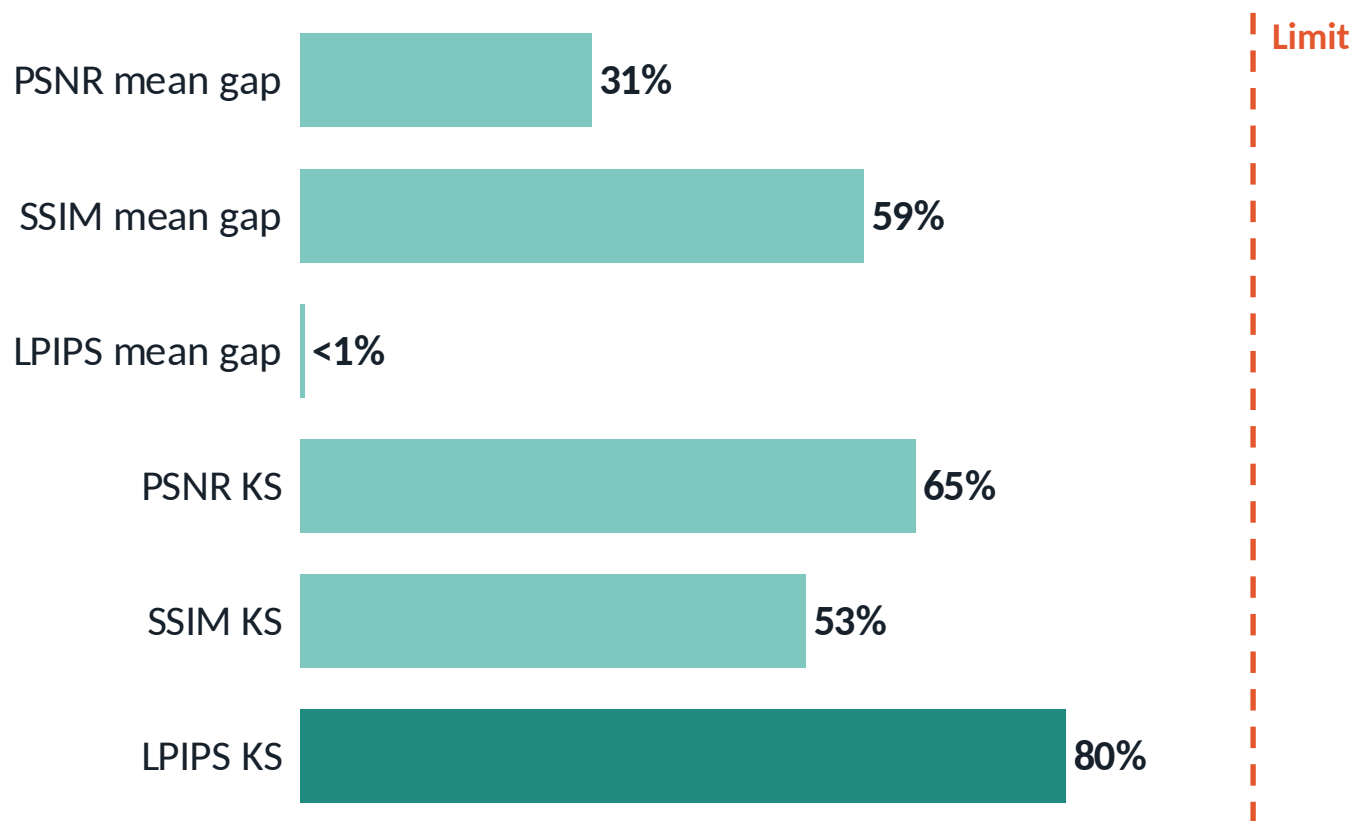
RASS-96

96 scenes · LCB 0.241

Compact reporting

RASS-48 passes the global audit with 73× fewer scenes

Share of each tolerance that RASS-48 uses



73×

fewer scenes: 3,521 → 48

8 scenes from each of the
6 descriptor regimes



Joint pass, all six

One audit covers more methods, generators, and datasets



RASS-96 passes the 3- and 4-method audits

Constraints passed in joint audits

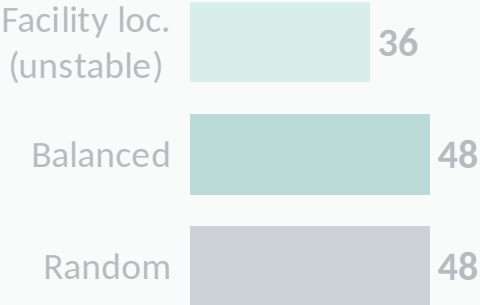
	3 methods	+ nerfacto
RASS-48	14/16	19/22
RASS-96	16/16 ✓	22/22 ✓

So RASS-96 is our subset for reported results.



Any candidate generator plugs in

Smallest certified size (scenes)



✗ k-center fails up to 96

Balanced also passes at 36 on an extended grid.



The protocol transfers to DL3DV

64 / 140

DL3DV scenes certified for four methods (32 for one)

Contract re-declared for DL3DV; proportional allocation, added post hoc.

One audit covers more methods, generators, and datasets



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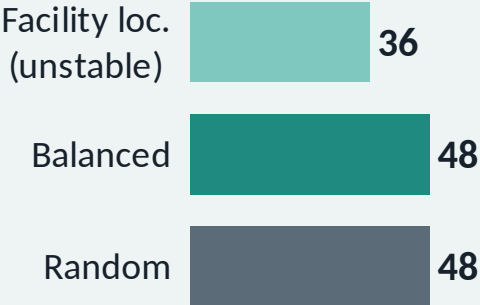
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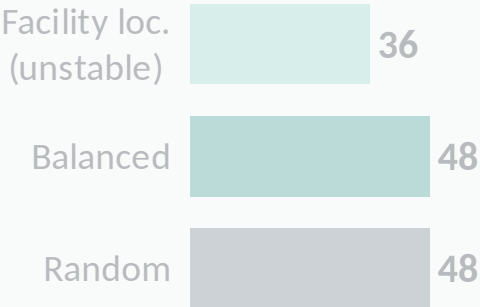
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Certify the procedure, not one lucky subset

RASS-48

Screening & ablations

Smallest evaluated budget with $LCB \geq 0.08$

RASS-96

Compact reporting

$LCB \geq 0.20$; passes the 3- and 4-method joint audits

How to use: train only on RASS-48 or RASS-96, compare with our released logs on the same scenes, and report the subset version.



github.com/GCVCG/RASS



kaggle.com/datasets/nobodyeh/rass-nerf-benchmark-artifact



Code & data

Thank you